

Evaluating Limits Worksheet — Worked Answer Key

Twenty-four limits that progress from substitution to algebraic, one-sided, and infinite behavior.

Revision: 2026-07-23 **Canonical page:** <https://bettergrades.net/subjects/math/calculus/worksheets/evaluating-limits/>

1. Evaluate $\lim_{x \rightarrow 2} (3x^2 - 5x + 4)$.

Answer: 6

- The expression is continuous at the target input 2.
- Substitute the target and simplify to obtain 6.

2. Evaluate $\lim_{x \rightarrow -2} (x^3 + 4x)$.

Answer: -16

- The expression is continuous at the target input -2.
- Substitute the target and simplify to obtain -16.

3. Evaluate $\lim_{t \rightarrow 3} \frac{t^2 + 1}{t + 2}$.

Answer: 2

- The expression is continuous at the target input 3.
- Substitute the target and simplify to obtain 2.

4. Evaluate $\lim_{h \rightarrow 0} (7 - 4h + h^2)$.

Answer: 7

- The expression is continuous at the target input 0.
- Substitute the target and simplify to obtain 7.

5. Evaluate $\lim_{u \rightarrow 1} \sqrt{u + 8}$.

Answer: 3

- The expression is continuous at the target input 1.
- Substitute the target and simplify to obtain 3.

6. Evaluate $\lim_{x \rightarrow 4} \frac{x+5}{\sqrt{x}}$.

Answer: $\frac{9}{2}$

- The expression is continuous at the target input 4.
- Substitute the target and simplify to obtain $\frac{9}{2}$.

7. Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$.

Answer: 6

- Factor the vanishing numerator as $(x - 3)(x + 3)$.
- Cancel the common factor only for nearby inputs, then substitute the target.
- The simplified expression approaches 6.

8. Evaluate $\lim_{x \rightarrow -4} \frac{x^2 - 16}{x + 4}$.

Answer: -8

- Factor the vanishing numerator as $(x + 4)(x - 4)$.
- Cancel the common factor only for nearby inputs, then substitute the target.
- The simplified expression approaches -8 .

9. Evaluate $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$.

Answer: 12

- Factor the vanishing numerator as $(x - 2)(x^2 + 2x + 4)$.
- Cancel the common factor only for nearby inputs, then substitute the target.
- The simplified expression approaches 12 .

10. Evaluate $\lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x - 1}$.

Answer: 3

- Factor the vanishing numerator as $(x - 1)(x + 2)$.
- Cancel the common factor only for nearby inputs, then substitute the target.
- The simplified expression approaches 3 .

11. Evaluate $\lim_{y \rightarrow 5} \frac{y^2 - 7y + 10}{y - 5}$.

Answer: 3

- Factor the vanishing numerator as $(y - 5)(y - 2)$.
- Cancel the common factor only for nearby inputs, then substitute the target.
- The simplified expression approaches 3 .

12. Evaluate $\lim_{z \rightarrow -1} \frac{z^3 + 1}{z + 1}$.

Answer: 3

- Factor the vanishing numerator as $(z + 1)(z^2 - z + 1)$.
- Cancel the common factor only for nearby inputs, then substitute the target.
- The simplified expression approaches 3 .

13. Evaluate $\lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x - 9}$.

Answer: $\frac{1}{6}$

- Multiply numerator and denominator by the conjugate $\sqrt{x} + 3$.
- Use the difference-of-squares identity and cancel the factor that tends to zero.
- Substitution in the simplified expression gives $\frac{1}{6}$.

14. Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$.

Answer: $\frac{1}{2}$

- Multiply numerator and denominator by the conjugate $\sqrt{1+x} + 1$.
- Use the difference-of-squares identity and cancel the factor that tends to zero.

c. Substitution in the simplified expression gives $\frac{1}{2}$.

15. Evaluate $\lim_{t \rightarrow 4} \frac{t-4}{\sqrt{t}-2}$.

Answer: 4

a. Multiply numerator and denominator by the conjugate $\sqrt{t} + 2$.

b. Use the difference-of-squares identity and cancel the factor that tends to zero.

c. Substitution in the simplified expression gives 4.

16. Evaluate $\lim_{u \rightarrow 0} \frac{u}{\sqrt{9+u}-3}$.

Answer: 6

a. Multiply numerator and denominator by the conjugate $\sqrt{9+u} + 3$.

b. Use the difference-of-squares identity and cancel the factor that tends to zero.

c. Substitution in the simplified expression gives 6.

17. For $f(x) = \begin{cases} x+2, & x < 1 \\ 4-x, & x \geq 1 \end{cases}$, evaluate the limit as $x \rightarrow 1^-$.

Answer: 3

a. Select the branch determined by the direction of approach.

b. Evaluate nearby behavior on that branch; the value assigned at the endpoint does not control the limit.

c. The required approach gives 3.

18. For $f(x) = \begin{cases} 2x, & x < 0 \\ x^2 + 1, & x \geq 0 \end{cases}$, evaluate the limit as $x \rightarrow 0^+$.

Answer: 1

a. Select the branch determined by the direction of approach.

b. Evaluate nearby behavior on that branch; the value assigned at the endpoint does not control the limit.

c. The required approach gives 1.

19. For $g(x) = \frac{|x|}{x}$, evaluate the limit as $x \rightarrow 0^-$.

Answer: -1

a. Select the branch determined by the direction of approach.

b. Evaluate nearby behavior on that branch; the value assigned at the endpoint does not control the limit.

c. The required approach gives -1.

20. For $h(x) = \begin{cases} x^2, & x \leq 2 \\ 6-x, & x > 2 \end{cases}$, evaluate the limit as $x \rightarrow 2$.

Answer: 4

a. Select the branch determined by the direction of approach.

- b. Evaluate nearby behavior on that branch; the value assigned at the endpoint does not control the limit.
- c. The required approach gives 4.

21. Evaluate $\lim_{x \rightarrow \infty} \frac{5x^2-1}{2x^2+3}$.

Answer: $\frac{5}{2}$

- a. Identify the dominant behavior: equal degrees.
- b. Divide by the highest relevant power or use the sign of the vanishing factor.
- c. The limit is $\frac{5}{2}$.

22. Evaluate $\lim_{x \rightarrow \infty} \frac{3x+7}{x^2+1}$.

Answer: 0

- a. Identify the dominant behavior: denominator degree is larger.
- b. Divide by the highest relevant power or use the sign of the vanishing factor.
- c. The limit is 0.

23. Evaluate $\lim_{x \rightarrow -\infty} \frac{4x^3+x}{2x^3-5}$.

Answer: 2

- a. Identify the dominant behavior: equal degrees.
- b. Divide by the highest relevant power or use the sign of the vanishing factor.
- c. The limit is 2.

24. Evaluate $\lim_{x \rightarrow 2^+} \frac{1}{x-2}$.

Answer: $+\infty$

- a. Identify the dominant behavior: positive denominator approaching zero.
- b. Divide by the highest relevant power or use the sign of the vanishing factor.
- c. The limit is $+\infty$.